

WEEK 5



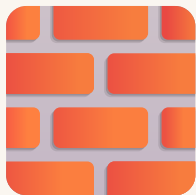
# Mini Cube Puzzle

Build it, then break out the top

큐브를 쌓고 윗면을 뚫고 나오기 (영어코딩)

BIG PICTURE

# From a path to a structure



Loops + conditionals place blocks, then carve an exit.

반복문과 조건문으로 큐브를 짓고 빠져나와요

## HOW TO COUNT

# Count the axis — one block, one number



- 1 Stand on the start block — that spot is  $(0, 0, 0)$ .
- 2  $x \rightarrow$  walk **right**. Add **+1** for every block.
- 3  $y \rightarrow$  jump **up**. Add **+1** for every block.
- 4  $z \rightarrow$  walk **forward**. Add **+1** for every block.

Say them in order every time:  $(x, y, z) = \text{across, up, forward}$ .

한 칸 움직이면 +1 — 오른쪽  $x$ , 위로  $y$ , 앞으로  $z$ . 순서는 늘  $(x, y, z)$

TYPE IT IN

# Put the numbers into Minecraft



<sup>x</sup>  
across →

<sup>y</sup>  
up ↑

<sup>z</sup>  
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

## Place + move = a line of blocks

```
for i in range(5):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

Drop a block, step forward, repeat 5× → a 5-long wall.

블록 놓고 한 칸 이동하기를 반복

## CONCEPT

# A conditional decides each spot

```
if on_edge or on_top_layer:  
    agent.place(DOWN)  
else:  
    # leave it empty (air)  
    pass
```

Edges and the top get a block; the inside stays hollow.

가장자리와 윗면만 블록을 놓아요

CODE

## Fill a layer, then go up

```
for layer in range(5):  
    fill_layer()  
    agent.move(UP, 1)
```

Finish the **whole** layer *before* moving up — order matters.

한 층을 다 채운 뒤에 위로 올라가요

CODE

## Break a hole and climb out

```
# make the hole FIRST, then move into it  
agent.destroy(UP)  
agent.move(UP, 1)
```

Carve the block above, **then** climb. Never the other way around.

먼저 구멍을 뚫고 그다음에 올라가요



PREDICT 🤖

# What shape appears?

Predict first — then click reveal

```
for i in range(5):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

Reveal output 🔍 reveal

VOCAB

# New words today

layer

one flat level of the cube

층

place

put a block where the agent points

놓다

destroy

break the block in a direction

부수다

edge

the outer border of a layer

가장자리



# What's wrong with this code?

Goal: finish the layer, then move up.

buggy

```
agent.place(DOWN)
agent.move(UP, 1)
agent.place(DOWN)
agent.place(DOWN)
```

Think first — then click next.

 DEBUG

# The bug

**Bug:** the agent moves up after only one block, leaving the layer unfinished. Place *all* the layer's blocks first, *then* move up once.

fixed

```
agent.place(DOWN)
agent.place(DOWN)
agent.place(DOWN)
agent.move(UP, 1)
```

YOUR TURN!

## Finish the cube

- ✓ Open the Week 5 world — find the cube starter
- ✓ Keep the pattern going (Steady) or add 2 conditionals (Challenge)
- ✓ Make one of them use `or`
- ✓ Break a hole in the top and climb out
- ✓ Send a photo or video on KakaoTalk



## QUIZ

QUESTION 1 OF 3

What does `for i in range(5)` do?

Only turns the agent.

Runs the body once.

Runs forever.

Runs the body 5 times.



## QUIZ

QUESTION 2 OF 3

Why finish a layer before `agent.move(UP, 1)` ?

Moving up early leaves the current layer unfinished.

The agent can't move up otherwise.

It places blocks faster.

It doesn't matter at all.



## QUIZ

QUESTION 3 OF 3

To climb out the top, what order is correct?

Move UP first, then destroy.

Destroy UP first, then move UP into the hole.

Turn right, then move down.

Place a block UP, then move.



QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



# Cube complete!

Next: flip levers to open redstone doors.

다음 주: 레버로 레드스톤 문 열기